

The Weight Norm Sets the Grokking Timescale: A Causal Delay Law

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Abstract

We show that the weight norm causally controls the *timescale* of grokking, reconciling two opposing accounts: that grokking occurs at a critical weight norm, and that the norm is not the operative variable. On modular arithmetic, a network first memorizes by raising its weight norm and then relaxes under weight decay; under free dynamics generalization emerges as the norm reaches a sharply concentrated value $\|W\|_c$ (1–2% across a two-fold range of training fractions, nearly unchanged across a ten-fold range of learning rates), matching recent threshold reports. We then intervene: a matched-counterfactual clamp holding $\|W\| = \rho \|W\|_c$ throughout training shows the network groks *at whatever norm it is held at* — no single norm is required — with the time to grok growing as an exponential delay law $T_{\text{grok}} \propto e^{\alpha\rho}$ ($R^2 > 0.99$). Across five tested moduli (four usable for the fit) this is a *scaling law*: one shared exponent $\alpha \approx 7.5$ collapses the delay ($R^2 = 0.996$), the norm entering only through its value relative to $\|W\|_c(p)$. Holding the norm above $\|W\|_c$ thus delays grokking rather than preventing it — an apparent “prevention” under a fixed budget is the finite-budget tail of this law — and the norm dominates the grokking time ($\approx 19\times$) over the learning rate ($\approx 2\times$). This reconciles the camps: the concentrated $\|W\|_c$ is the norm the relaxation *reaches* on its natural timescale, not a hard gate, so grokking is seen above and below it; the result is a causal complement to norm-separation delay theory, whose closed-form delay is logarithmic in the norm ratio under free contraction. The effect is altered by *LayerNorm*, which decouples the weight-norm scale from the function. On an un-normalized attention model the delay law recurs with its own exponent ($\alpha_2 \approx 15$, $R^2 = 0.999$; $\approx 2\times$ the MLP), so the *form* of the law is cross-architectural while its rate is not. A concentrated, regularization-set grokking norm also recurs on a non-Fourier task (sparse parity), although the above-norm delay does not fully transfer.

Keywords: grokking; delayed generalization; weight

norm; critical norm; causal intervention; weight decay; delay law; scaling law; mechanistic interpretability; learning dynamics.

1 Introduction

Grokking — delayed generalization long after a network has fit its training set — has become a testbed for understanding when and why neural networks generalize [1]. Two influential accounts have emerged. The *weight-norm* account, exemplified by *Omnigrok* [6], holds that generalization is governed by the parameter norm: small-norm regions generalize and weight decay drives the network there. The *circuit* account [2] shows mechanistically that the network gradually builds Fourier-feature “circuits” that implement the arithmetic. These have largely been studied separately and qualitatively.

This paper makes the weight-norm account *sharp, quantitative, and causal*, and connects it to the circuit account. The literature is currently divided: one line reports a concentrated weight-norm threshold for grokking [6, 18, 19], while another argues the norm is correlative or not the operative variable at all [16, 17, 20, 21]. Our central aim is to reconcile this disagreement. We record the weight norm at grokking under free dynamics and find it concentrated at $\|W\|_c$ (in agreement with the threshold camp); we then intervene with a matched-counterfactual clamp that holds the norm at a chosen multiple of $\|W\|_c$ throughout training, and find that the norm does not gate grokking at a fixed value but sets its *timescale* — a causal delay law. This dissolves the disagreement: $\|W\|_c$ is the norm the free relaxation reaches on its natural timescale, not a value grokking requires. This is also consistent with a possible division of labor between the two accounts — circuit formation governing *what* is learned and norm relaxation *when* it generalizes — though our evidence for the link is the temporal alignment of the two, not a direct manipulation of the circuits.

Contributions.

1. **A concentrated grokking norm under free dynamics.** Trained freely, the model groks when the weight norm reaches a sharply concentrated value $\|W\|_c$: for fixed task size and weight decay, $\|W\|_c$ varies by only 1–2% across a two-fold range of training fractions and is nearly invariant across a ten-fold range of learning rates (§4). Concurrent studies report the same rate-vs-threshold dissociation [18, 19]; we corroborate it and then show, causally, that it is a *rate* phenomenon.
2. **The weight norm causally sets the grokking timescale (a delay law).** A matched-counterfactual clamp that holds $\|W\| = \rho \|W\|_c$ throughout training reveals three things (§5): (i) the network groks at *whatever* norm it is held at — there is no single norm grokking requires; (ii) over $\rho \in [0.85, 1.15]$ the time to grok follows a clean exponential delay law $T_{\text{grok}} \propto e^{\alpha\rho}$ ($R^2 > 0.99$); (iii) holding the norm above $\|W\|_c$ therefore *delays* grokking rather than preventing it — an apparent “prevention” under a fixed budget is the finite-budget tail of this exponential. The norm is the dominant control of grokking time ($\approx 19\times$ across the range), with the learning rate a secondary, dissociable modulator ($\approx 2\times$).
3. **The delay is a scaling law with a shared exponent across task sizes.** Repeating the clamp across five tested moduli p , the four usable sizes collapse onto a single exponential in the *relative* norm $\rho = \|W\|/\|W\|_c(p)$: one shared exponent $\alpha \approx 7.5$ fits them with $R^2 = 0.996$ (per- p CV 4.1% over the tested $1.7\times$ size range), establishing a causal scaling law rather than a per-task fit (§6). We report its limits honestly — sub-exponential saturation beyond $\rho \approx 1.25$ and a minimum-norm floor at the smallest p (which is why it is excluded from the fit) — and relate it to norm-separation delay theory [28]: that theory predicts a *logarithmic* free-training delay via norm contraction, whereas our clamp blocks contraction and is a continuous-dose version of the norm-freeze causal test, confirming the norm as the operative variable (we do not claim a quantitative match of the exponent to a closed-form prediction).
4. **Recurrence across architectures, and the role of normalization.** The delay law recurs on a second, un-normalized attention model with its own exponent ($\alpha_2 \approx 15$, $R^2 = 0.999$; grok-at-held-norm and high-norm saturation as on the MLP), and is altered by LayerNorm, which decouples the weight-norm scale from the function; only the functionally relevant unembedding norm stays concentrated. The weight norm controls the timescale when it sets the function scale (§7).

5. **Task generality on a non-Fourier task.** On sparse parity, grokking again occurs at a concentrated, weight-decay-set, learning-rate-invariant norm, and the norm again acts through the rate rather than a hard threshold (§8).
6. **A temporal link to the circuit account.** Fourier-feature concentration rises in the same norm-relaxation window. This temporal alignment is consistent with — but does not establish — norm relaxation organizing the feature formation of the circuit account; we do not intervene on the circuits directly, and we make the connection only at the correlational level.

2 Related Work

Grokking: phenomenology and mechanism.

Grokking was identified by Power et al. [1] on small algorithmic tasks, where test accuracy rises abruptly long after the training set is fit. Nanda et al. [2] reverse-engineered the learned solution on modular arithmetic as a small set of Fourier-frequency circuits implementing a trigonometric identity, and introduced progress measures (e.g. a restricted loss) that improve smoothly before the visible jump; Gromov [3] gave analytic constructions of such solutions, and Barak et al. [4] showed on sparse parity that hidden progress accumulates throughout the plateau. Varma et al. [5] framed grokking as competition between a memorizing and a generalizing circuit of differing parameter efficiency, with weight decay tipping the balance. These works establish *what* the network computes and that progress is gradual; our focus is the scalar control variable — the weight norm — and its causal role in triggering the transition.

A concentrated weight-norm threshold: an emerging, but purely observational, consensus.

A second line attributes grokking to the parameter norm and the optimizer’s implicit bias. Liu et al. [6] (Omnigrok) argued that generalization is organized by the weight norm: a generalizing region exists at moderate norm, weight decay or rescaled initialization moves the network into it, and with weight decay as the only regularizer the norm relaxes as $\|w(t) \approx e^{-\gamma t} \|w_0\|$ toward a target w_c in time $\propto 1/\lambda$. Related accounts cast the jump as a lazy-to-rich transition [7], or as a late-phase implicit-bias dichotomy [8], connected to the classical max-margin bias of gradient descent [9]. Two concurrent empirical studies sharpen the role of regularization in a way closely related to ours, and we are careful to delineate what they establish from what we add. Manir & Rupa [18] run a factorial sweep over depth, architecture, activation, and weight decay on modular addition,

finding that grokking is driven by the interaction of optimization and regularization rather than by architecture — weight decay is the dominant control, with a narrow “Goldilocks” regime, and the Transformer–MLP gap nearly vanishes ($1.11\times$ delay) under matched hyperparameters; in a single weight-norm experiment (one seed per configuration) they report that the RMS parameter norm at grokking is concentrated to 0.0219 ± 0.0032 (CV 14.5%) across five width-512 models, with ReLU and GELU grokking at *identical* norms despite a $6.9\times$ difference in delay. A parallel study of neural-collapse dynamics [19] reports the same rate-vs-threshold dissociation for a *different* observable — the penultimate-layer *feature* norm in image classification — with a concentrated onset value (within-pair CV $< 8\%$) that is largely invariant to training conditions while training sets only the *rate* of approach, again with a Goldilocks zone. Both studies are *observational* on the norm–time relation: each records the norm at which the transition occurs and varies hyperparameters around it; neither holds the norm fixed, measures a delay as a function of a controlled norm, or reports a scaling law. Our observational findings (§4) corroborate this rate-vs-threshold dissociation and tighten the concentration to CV 1–2% by conditioning on task and λ , adding invariance to learning rate and to the grokking threshold, and a modulus scaling $\|W\|_c \propto p^{0.38}$. We regard the concentrated threshold itself as a *corroborated* phenomenon rather than a novel one; our distinct contribution begins where these studies stop — we test the threshold *causally* by clamping the norm and reading off the resulting delay (§5), and show the response is a quantitative, cross-task delay law (§6) rather than another observational characterization. (The cross-architecture comparison above is at the natural operating point; our $\alpha_2 \approx 2\alpha$ for the un-normalized transformer measures a different quantity, the sensitivity of the delay to a *held* norm, so the two are consistent.)

Challenges to weight-norm causality. A vigorous counter-current argues the weight norm is *not* the operative variable. Golechha [16] reports grokking on real-world data (MNIST, IMDB) outside the expected norm range — even without weight decay, with the norm increasing — concluding weight norms are correlative, not causal. Minegishi et al. [17] show a sparse “grokking ticket” generalizes faster than a *norm-matched* dense network, so the norm is not sufficient. Notsawo et al. [20] demonstrate that grokking can be driven by regularization toward properties other than the ℓ_2 norm (sparsity, low rank), that the ℓ_2 norm is not a reliable proxy — it often grows while the model generalizes — and that depth alone can induce grokking without explicit regularization. Lyle et al. [21] argue that what drives the transition is the *effective learning*

rate — the ratio of parameter norm to update norm — rather than the parameter norm on its own, and show that deliberately raising it accelerates feature learning (and mitigates primacy bias under nonstationarity); further work induces grokking with increasing norms and no weight decay [22]. We take this critique seriously and do not claim the norm is universally causal. Instead, our results delimit *when* the norm is causal: it controls the grokking timescale in settings where it sets the function scale, and that control weakens or disappears exactly where these works operate — under normalization (§7), when regularization targets a non- ℓ_2 property, or on a task whose solution is not pinned to a single norm (§8). To separate the norm *state* from the effective learning rate, our intervention holds the norm fixed while the optimizer runs (without resetting AdamW moments) and measures a dose-response across clamp levels at fixed learning rate (§5).

Interventions on the norm, and the role of normalization. The interventional evidence closest to ours comes from two different settings, and in each the intervention differs from our sustained clamp in a way that turns out to be decisive. In the neural-collapse study, Rupa [19] rescales the penultimate *feature* norm to $0.3\times$ or $3.0\times$ its natural value at the start of the collapse phase and then releases it; the feature norm self-corrects back to the same value from either direction, with collapse time essentially unchanged ($p > 0.2$), and the authors read this as evidence that the threshold is a gradient-flow *attractor rather than a cause*. This is a one-shot rescale-and-release, and its null result is the same one we obtain for the grokking weight norm in our own one-shot control (§5, “Sustained, not instantaneous”): a single rescale washes out because the optimizer re-equilibrates the norm, leaving the grokking time within $\sim 15\%$ of the free control. Far from preempting our finding, that null result *motivates* our design — the effect appears only under a *sustained* clamp that holds the norm away from its attractor throughout training, which is what produces the exponential dose-response. The two studies are thus complementary across both domain (feature norm in neural collapse vs. weight norm in grokking) and intervention type (transient rescale vs. sustained hold). Closest to our weight-norm setting, Verma [23] runs a matched-control intervention on a LayerNorm transformer: re-initializing attention heads changes post-grokking head differentiation, while a *matched weight-norm clip* does not, isolating the effect to head structure rather than weight magnitude. This null result for norm-clipping is consistent with — and explained by — our finding that LayerNorm decouples the weight-norm scale from the function, so that clamping the total norm should have little effect there; our gate appears precisely in the *un-normalized*

setting (§7). The decoupling itself has a formal basis: for a final normalization layer the radial gradient at the output vanishes, so the loss cannot be reduced by rescaling the last hidden state, whereas without it cross-entropy pushes representation norms up [24]; high-norm directions act on logits only through the final LayerNorm [25], and LayerNorm’s scale-invariance is classical [26]. Concurrently, Yildirim [27] relates normalization, the unembedding matrix, and logit scale to grokking onset via a bounded-geometry intervention. Our contribution in this setting is to show which norm remains concentrated under LayerNorm (only the functionally relevant unembedding norm) and that the norm’s control of the grokking timescale reappears once normalization is removed — a grokking-specific instantiation of the general decoupling these works establish.

Relationship to our prior work. A companion theoretical paper from our group derives a norm-separation delay law from first principles, predicting that the *free-training* grokking delay grows logarithmically in the norm ratio as the norm contracts to threshold [28]. The present paper differs in object and method: it intervenes directly on the scalar weight-norm *state* (rather than deriving free dynamics) and maps the conditions under which that state is causal for the onset of grokking. Where that theory predicts the free-training delay as logarithmic in the norm ratio, our pinned-norm intervention measures the complementary regime in which contraction is blocked, finding an exponential dependence on the held relative norm. We build on, and cite, that work rather than restating it.

Grokking and emergence as phase transitions. Abrupt capability jumps in training and at scale invite a statistical-physics reading. Emergent abilities [11] were argued by Schaeffer et al. [12] to be partly an artifact of discontinuous metrics, sharpening the need for well-defined order parameters and finite-size analysis; singular learning theory treats learning as passing through phases of differing degeneracy [13]. Within this view our contribution is methodological as well as empirical: we identify a scalar control parameter — the weight norm — and measure its causal, quantitative effect on the timescale of the transition, treating grokking as a phenomenon to be measured rather than merely analogized.

Double descent. Epoch-wise and model-wise double descent [14, 15] are distinct generalization transitions in which test error is non-monotone near the interpolation threshold. We use double descent as a contrast (§9): in a linear model it is analytically a saddle-node bifurcation with a different critical exponent, placing it in a different universality class from the grokking transition studied here.

3 Setup

Task and data. We study modular addition $f(a, b) = (a + b) \bmod p$ for prime p . Each example is the token sequence $[a, b, =]$ over a vocabulary of size $p+1$ (the p residues plus an “=” token), with target the residue $f(a, b)$. The dataset is the full set of p^2 pairs; a fraction α is sampled per seed as the training set and the remainder held out. The training fraction α is our control parameter; the modulus p (dataset size p^2) is our system size.

Model. Unless noted, the network is a two-layer MLP over learned token embeddings: an embedding matrix $E \in \mathbb{R}^{p \times d}$, the concatenation of the two operand embeddings ($\in \mathbb{R}^{2d}$), a hidden layer $W_1 \in \mathbb{R}^{2d \times H}$ with GELU, and an output $W_2 \in \mathbb{R}^{H \times p}$; biases carry no weight decay. We use $d = 128$, $H = 256$. A one-layer multi-head attention model (token and positional embeddings, self-attention, an MLP block, and an unembedding, predicting at the final position) is used for the architecture study (§7), in two variants — with LayerNorm and without — to isolate the role of normalization.

Optimization. We train full-batch with AdamW ($\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$), learning rate r , and decoupled weight decay λ applied to weight matrices only; parameters are initialized at scale $1/\sqrt{\text{fan-in}}$. Reference hyperparameters are $r = 10^{-3}$, $\lambda = 1$ unless swept.

Observables. On a geometric grid of training steps and across 12–16 seeds we record train and test accuracy; the total weight norm

$$\|W\| = (\|E\|_F^2 + \|W_1\|_F^2 + \|W_2\|_F^2)^{1/2}, \quad (1)$$

the Omnigrok control variable; and the embedding’s Fourier concentration, the Gini coefficient of its token-frequency power spectrum $P_k = \sum_j |\hat{E}_{k,j}|^2$ (DC term removed, normalized to a distribution), which is low for an unstructured embedding and high once a few Fourier modes dominate. We define the memorization time T_{mem} as the first step with train accuracy ≥ 0.99 , the grokking time T_{grok} as the first step with test accuracy ≥ 0.9 (robustness to this threshold is examined in §4), and the critical norm $\|W\|_c = \|W\|$ at T_{grok} .

Modulus sweep. For the observational scaling of the critical norm (§4) we sweep $p \in \{29, 41, 59, 79, 97\}$.

Intervention (matched counterfactual). For the causal test (§5) the random seed is derived from the base configuration only, *excluding* the intervention, so the control and every intervention share identical initialization, data split, and pre-intervention trajectory. The norm *clamp* projects the weights after each step so that $\|W\| = \rho \|W\|_c$ for a chosen multiple ρ , holding the norm at a multiple of the critical value; a one-shot

variant applies the projection once at a fixed step t_{int} (after memorization, before grokking).

Reproducibility. All experiment runners write atomically with per-configuration resume and store raw trajectories and intermediate snapshots (including periodic embedding matrices) for later analysis; code and data are released.

4 The Critical Weight Norm

Recording $\|W\|$ at the grokking step reveals a sharp regularity: within a fixed architecture, optimizer, task size, and weight-decay strength, grokking occurs at a critical norm $\|W\|_c$ concentrated to within a few percent, varying only with weight decay and system size. We emphasize that this section is observational; its causal status is tested in §5.

Concentrated, and stable across data fraction and learning rate. At fixed (p, λ, r) , $\|W\|$ at grokking is nearly constant across the training fraction α : for $p = 59$ it is $\{56.4, 55.9, 54.8, 55.1, 55.7\}$ at $\alpha = \{0.30, 0.35, 0.40, 0.50, 0.65\}$ — a coefficient of variation of 1–2% across a two-fold range of data (Fig. 1A). At sufficient weight decay it is also nearly independent of the learning rate: at $\lambda = 3$, $\|W\|_c = \{48.9, 48.7, 49.1\}$ across a ten-fold range of r (CV < 1%), even though the grokking time changes seven-fold. The learning rate thus changes the *time to reach* $\|W\|_c$ much more than the value of $\|W\|_c$, separating a tunable relaxation rate from a comparatively stable threshold; this is consistent with the $(r\lambda)$ -dependence of the grokking time.

An internal threshold, not a training-time artifact. $\|W\|_c$ is robust to the accuracy threshold defining grokking: recomputing it at test-accuracy thresholds 0.8, 0.9, 0.95 changes $\|W\|_c$ by less than 2% (e.g. $p = 59$: 55.0/54.6/54.2; $p = 97$: 68.0/67.1/66.4), with the data-fraction stability unchanged at every threshold. Because accuracy rises steeply at the transition while the norm relaxes slowly, the norm barely moves over the accuracy window 0.8–0.95: $\|W\|_c$ marks an internal state, not the placement of a threshold.

Set by weight decay; scales with system size. $\|W\|_c$ is not a universal constant. It shifts with weight decay — $\|W\|_c \approx 49, 55, 63$ at $\lambda = 3, 1, 0.3$, consistent with a weight-decay equilibrium norm — and scales with system size as $\|W\|_c(p) = \{42, 48, 55, 61, 67\}$ for $p = \{29, 41, 59, 79, 97\}$, a power law $\|W\|_c \propto p^{0.38}$ (Fig. 1B). The claim is one of threshold structure within a fixed $(p, \lambda, \text{architecture})$ family, not of a single universal value.

Temporally aligned with Fourier-feature formation. The norm overshoots during memorization (peaking near step 340 in Fig. 1C), after which weight decay relaxes it; grokking occurs on the descent as the norm approaches $\|W\|_c$. The embedding’s Fourier concentration rises in the same window (norm overshoot $\rightarrow$ feature formation $\rightarrow$ grokking), suggesting that norm relaxation toward $\|W\|_c$ *may organize* the feature formation of the circuit account. The trajectory shows temporal alignment; the next section establishes the causal role of the norm.

5 A Causal Test: the Weight Norm Sets the Grokking Timescale

The correlations above admit two readings. Either the norm *controls* grokking, or it is merely a by-product of the same training dynamics. We discriminate them with a controlled intervention. The answer is sharper than either the threshold or the no-effect view: the norm controls *when* grokking happens, through a clean delay law.

Design. We use a matched counterfactual: the initialization, data split, and pre-intervention trajectory are identical across conditions (the random seed depends only on the base configuration, not on the intervention or the learning rate), so any difference is attributable to the intervention alone. The intervention *clamps* the weight norm: at every step after a fixed point t_{int} (chosen after memorization, before grokking) we project the weights so that $\|W\| = \rho \|W\|_c$, holding the norm at a chosen multiple of the critical value without resetting the optimizer moments. To separate the norm *state* from the learning rate, we run a full $\rho \times \eta$ matrix, $\rho \in \{0.85, 1.00, 1.15, 1.30\}$ and learning rate $\eta \in \{1, 2, 4, 8\} \times 10^{-3}$, 16 seeds each, at $p = 59$, $\alpha = 0.40$, $\lambda = 1$, with a generous 30,000-step budget so that slow conditions are observed to completion rather than censored.

Result: the network groks at the held norm, after an exponentially growing delay. Two facts settle the causal question (Fig. 2, Table 1). First, the network groks at *whatever* norm it is held at: across $\rho \in \{0.85, 1.00, 1.15, 1.30\}$ the norm at grokking equals the clamp value (46.4, 54.6, 62.8, 71.0), not a fixed $\|W\|_c$. There is no single norm that grokking requires. Second, over the range $\rho \in [0.85, 1.15]$ the time to grok grows as a clean exponential in the held norm,

$$T_{\text{grok}}(\rho) \propto e^{\alpha\rho}, \quad \alpha \approx 7.4, \quad R^2 > 0.99, \quad (2)$$

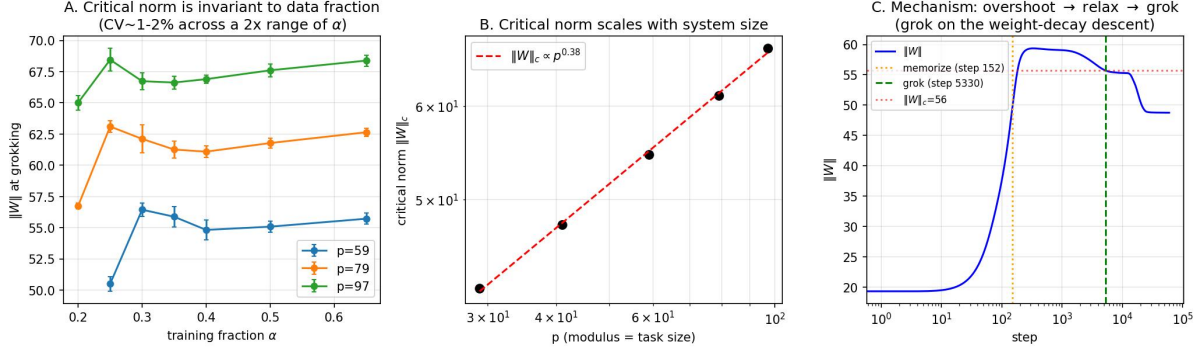


Figure 1: **A critical weight norm at grokking (observational).** (A) $\|W\|$ at grokking is concentrated and nearly invariant to the training fraction α (CV 1–2% across a 2x range of α). (B) It scales with system size, $\|W\|_c \propto p^{0.38}$. (C) The norm overshoots during memorization, then weight decay relaxes it; grokking occurs on the descent as $\|W\|$ approaches $\|W\|_c$, in the same window as Fourier-feature formation.

fit independently at each learning rate (Fig. 2A); a dense 10-level sweep below extends this exponential through $\rho = 1.25$ (Fig. 3), beyond which (at $\rho \approx 1.30$) the growth becomes sub-exponential (§6). Holding the norm above $\|W\|_c$ thus *delays* grokking rather than preventing it: at $\rho = 1.30$ ($\|W\| = 71$, 30% above critical) the median grokking time is $\approx 2.8 \times 10^4$ steps, so under a shorter budget the same condition appears to “never” generalize. Indeed no $\rho = 1.30$ seed groks before 20,000 steps, which is exactly why a fixed 20,000-step budget (and our own earlier report) read this as prevention; by 30,000 steps 12–16/16 seeds grok. The “prevention” is the finite-budget tail of the delay law, not a separate phenomenon.

Norm dominates; learning rate modulates; they dissociate. The norm is by far the larger lever: across the tested range it changes T_{grok} by $\approx 19\times$ (at $\eta = 10^{-3}$, T_{grok} rises from 1492 at $\rho = 0.85$ to 27,925 at $\rho = 1.30$). The learning rate changes it only $\approx 2\times$ (e.g. at $\rho = 1.30$, $27,925 \rightarrow 16,893$ as η goes $1 \rightarrow 8 \times 10^{-3}$). The two axes are separable: raising η shifts the delay-law intercept slightly but leaves the steep ρ -dependence intact (the curves in Fig. 2A are nearly parallel). The norm sets the *timescale* of grokking; the learning rate sets the rate within it. This directly addresses the view that grokking is governed by the effective learning rate rather than the norm [21]: the dominant, order-of-magnitude control here is the norm state, with the learning rate a secondary modulator.

A dense sweep confirms the exponential and locates its edge. To check that the exponential is not an artifact of fitting only three norm levels, we ran a fine sweep at $p = 59$ over ten clamp levels $\rho \in \{0.85, 0.90, \dots, 1.30\}$ (16 seeds each), measuring

Table 1: The weight norm causally sets the grokking *timescale* (matched control; $p = 59$, 16 seeds, 30,000-step budget). Median T_{grok} (steps) over the $\rho \times \eta$ clamp matrix; the network groks at the held norm in every cell. Bootstrap 95% CIs on the median are tight (e.g. $\rho=1.00, \eta=1e-3$: 4077 [3795, 4257]; $\rho=1.15, \eta=1e-3$: 13426 [13045, 13817]); per-cell CIs are in the supplement. All cells grok 16/16 within budget except the $\rho=1.30$ row at $\eta \in \{1, 4\} \times 10^{-3}$ (12–14/16; the remaining seeds are right-censored at 30,000, so those two medians are mild underestimates), which is why §6 re-runs $\rho=1.30$ at a 160,000-step budget where all seeds grok uncensored. The apparent “prevention” at high ρ in earlier short-budget runs is the finite-budget tail of the exponential delay (2).

held $\ W\ = \rho\ W\ _c$	$\eta=1e-3$	$2e-3$	$4e-3$	$8e-3$
0.85 $\ W\ _c = 46.4$	1492	916	697	587
1.00 $\ W\ _c = 54.6$	4077	2891	1961	1256
1.15 $\ W\ _c = 62.8$	13426	12317	10221	8007
1.30 $\ W\ _c = 71.0$	27925	23168	19501	16893

$\|W\|_c = 54.5$ from a free control. The network groks at the held norm in every cell, all 16/16 seeds, uncensored (Fig. 3). Over the nine levels $\rho \in [0.85, 1.25]$ the delay is a clean exponential, $\alpha = 7.64$ with a bootstrap 95% CI of [7.47, 7.81] and $R^2 = 0.994$ — a tight fit over a $30\times$ range of grokking times, not an underdetermined three-point line. The exponential regime is wider than our initial estimate: it holds through $\rho = 1.25$, and only at $\rho = 1.30$ does the delay fall clearly below the extrapolation ($\approx 0.69\times$, matching the p -scan value of §6), marking the onset of the sub-exponential saturation. There is a mild downward curvature at the very bottom ($\rho \leq 0.90$), where the held norm approaches the

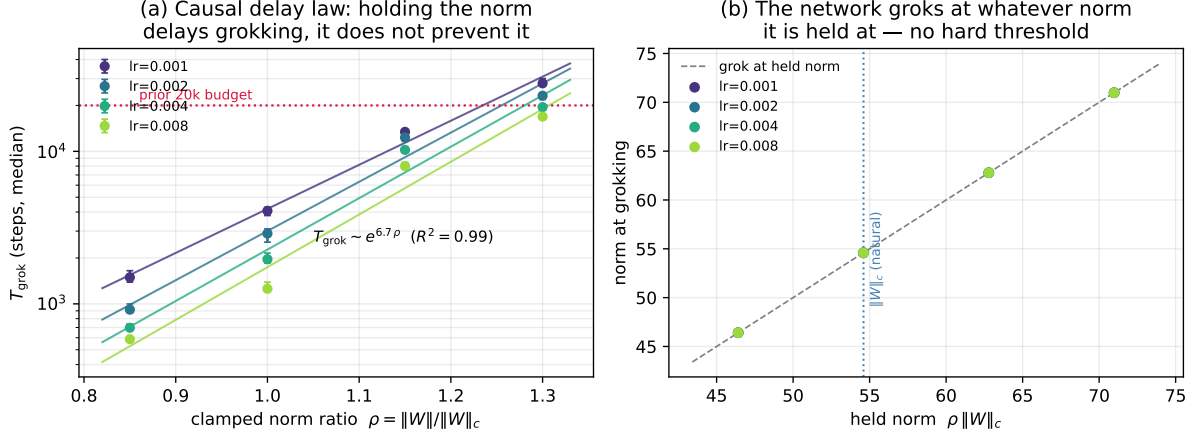


Figure 2: **The weight norm causally sets the grokking timescale.** (A) Holding the norm at $\rho\|W\|_c$ delays grokking as a clean exponential $T_{\text{grok}} \propto e^{\alpha\rho}$ ($R^2 = 0.99$); curves at four learning rates are nearly parallel (norm sets the timescale, η the rate). The dotted line marks a typical 20,000-step budget: high-norm clamps fall above it and look “prevented” under a short budget, but grok by 30,000 steps. (B) The network groks at *whatever* norm it is held at (points lie on the diagonal), so there is no single norm grokking requires; $\|W\|_c$ is merely the norm the free relaxation reaches.

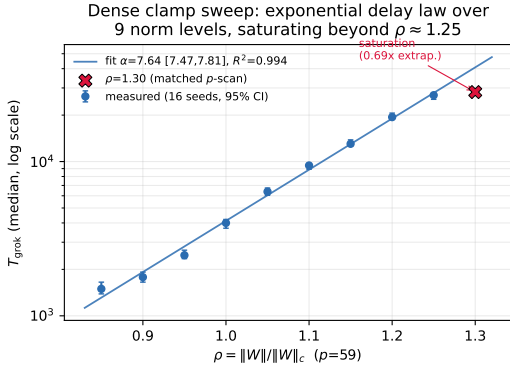


Figure 3: **Dense clamp sweep** ($p = 59$, 16 seeds). The grokking delay is exponential in the held relative norm over ten levels $\rho \in [0.85, 1.25]$ ($\alpha = 7.64$ [95% CI 7.47, 7.81], $R^2 = 0.994$); error bars are bootstrap 95% CIs on the median. The point at $\rho = 1.30$ (matched p -scan) falls below the extrapolation, marking the onset of sub-exponential saturation.

minimum functional norm (§6).

Is it the norm, or just stronger regularization?

A natural objection is that constraining the norm low is simply stronger regularization, which already accelerates grokking. We address this with a matched control. Training freely at a range of weight decays $\lambda \in \{1, 2, 3, 5, 8\}$ traces a regularization frontier on the $(\|W\| \text{ at grokking}, T_{\text{grok}})$ plane (lower equilibrium

norm, faster grokking). The below-clamp points lie *on* this frontier: the below-direction speed-up is not distinguishable from stronger regularization, and we do not claim it as a separate effect. The *above*-norm delay, however, is not reproducible by regularization: since weight decay can only *lower* the equilibrium norm, no free- λ run sits at the slow, high-norm states ($\|W\| = 63\text{--}71$) that the clamp holds, so the long delays at $\rho > 1$ are a property of the norm state that no weight-decay setting reaches. An *Omnigrok*-style low-norm initialization (rescaling the initial weights, then training freely) is washed out: the norm relaxes back toward $\|W\|_c$ and grokking occurs on the free timescale rather than being accelerated by the low start — consistent with the delay law, since only a *sustained* hold (not the initial value) changes the held norm the dynamics run at.

Sustained, not instantaneous. A *one-shot* rescale at a single step produces only a weak effect, because the optimizer re-equilibrates the norm and the kick washes out; the large effect requires the *continuous* clamp. Quantitatively, a one-shot global rescale to $\rho\|W\|_c$ that is then released leaves the grokking time nearly flat across $\rho \in [0.70, 1.30]$ (median T_{grok} moves only $1.2\times$, staying within $\sim 15\%$ of the free control), whereas a *sustained* hold of the same norm spans $7.3\times$ over the narrower range $\rho \in [0.90, 1.15]$. A one-shot $0.8\times$ scaling of any single parameter group (embedding, hidden, or output) likewise leaves T_{grok} within $\sim 3\%$ of the control. What matters is the norm state the network is held in throughout the relaxation, not a one-time crossing or

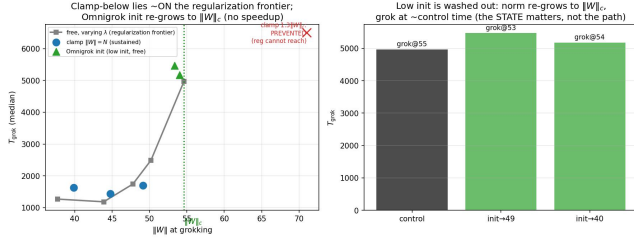


Figure 4: **The norm state, not just regularization.** (Left) Free training at varying weight decay traces a ($\|W\|$ at grok, T_{grok}) frontier; the below-direction norm clamps lie on it (their speed-up is regularization), but the slow high-norm states the clamp holds ($\rho > 1$) sit off the frontier, at norms no weight decay reaches, so the above-norm delay is a property of the norm state and not of regularization. (Right) An Omnigrok-style low-norm initialization is washed out: the norm relaxes back toward $\|W\|_c$ and grokking occurs on the free timescale, so only a *sustained* hold changes the timescale.

a transient perturbation of the optimizer — exactly as the delay law (2) requires.

The total norm, not inter-layer structure. The clamp is a strong, sustained constraint (a per-step projection in the spirit of Omnigrok’s rescaling), not a subtle perturbation. It rescales all weights by a single global factor, so it fixes the *total* magnitude while leaving the optimizer free to redistribute norm across layers. It does so: under the held clamp the per-layer norm fractions (embedding, hidden, output) still drift over training by as much (maximum shift ≈ 0.06) as in free training (≈ 0.06), even though the total is pinned. The delay is therefore a response to the total norm *value*, not to a freezing of the relative inter-layer structure. (We did not run a *sustained* per-layer pin, so we make no claim about how independently holding each layer’s norm would compare.) Overall the matrix establishes that the norm *state* causally sets the grokking timescale, dominant over the learning rate, and that holding the norm high delays rather than prevents grokking. The intervention does not by itself establish that norm relaxation causes the Fourier features specifically; that link remains temporal (§4).

6 A Scaling Law for the Grokking Delay

The single-task delay law (2) raises the question that separates a genuine law from a fitted curve: is the exponent α a property of one task size, or does the same

law govern grokking across task sizes? We answer this by repeating the clamp intervention across five moduli $p \in \{31, 41, 53, 59, 71\}$. For each p we first measure $\|W\|_c(p)$ from a free control and then clamp the norm at $\rho \|W\|_c(p)$ for $\rho \in \{0.85, 1.00, 1.15\}$ (and $\rho = 1.30$ at $p \in \{41, 59\}$), 12 seeds per cell. The smallest modulus $p = 31$ does not grok when clamped at $\rho = 0.85$ — the held norm falls below the minimum functional norm for the task (a floor we discuss in §6) — so we fit the exponent on the four larger moduli and treat $p = 31$ as the low-end boundary of the usable range.

A shared exponent across the tested task sizes.

Two results establish a scaling law (Fig. 5). First, at every task size the network again groks at *exactly* the held norm (norm at grokking = $\rho \|W\|_c(p)$ in every cell; each cell groks 12/12 seeds with the slowest seed well within budget, so none of these medians is censored), and the delay follows the same exponential form over $\rho \in [0.85, 1.15]$ with a shared exponent: fitting each p independently gives $\alpha = 7.92, 7.56, 7.42, 7.07$ for $p = 41, 53, 59, 71$ ($R^2 \geq 0.986$), a mean of $\bar{\alpha} = 7.49$ (coefficient of variation 4.1%). The point estimates decrease monotonically with p , but this drift is within seed noise: the bootstrap 95% CIs of the four per- p exponents all overlap (in $[7.06, 7.38]$), and replacing the single shared slope by four free per- p slopes (three extra parameters) reduces the residual sum of squares only from 0.054 to 0.037, which no model-selection criterion justifies. The data are therefore consistent with a single exponent. Second, the curves *collapse*: a global model $\log T_{\text{grok}} = \alpha \rho + f(p)$ with one shared exponent fits all four task sizes with a bootstrap estimate $\alpha = 7.49$ (95% CI $[7.16, 7.69]$) and $R^2 = 0.996$ (Fig. 5B). A single exponent across the tested $1.7\times$ range of p is the signature of a scaling law rather than a per-task coincidence: the weight norm sets the grokking timescale through the *same* exponential dependence on the norm *relative* to its task-specific critical value $\rho = \|W\|/\|W\|_c(p)$. (This relative normalisation is also the right one empirically: re-collapsing against the absolute norm $\|W\|$ degrades the fit to $R^2 = 0.988$, whereas ρ and other p -rescalings match within 0.001.) We still describe the exponent as shared over the *tested* finite-size range rather than proven universal, since four task sizes cannot exclude a slow drift outside this range. The critical value itself scales smoothly, $\|W\|_c(p) \propto p^{0.44}$, consistent with the observational scaling of §4, and the functionally relevant embedding norm scales similarly ($\|E\|_c \propto p^{0.47}$).

Honest limits of the exponential: high-norm saturation and a low-norm floor. The exponential is not unbounded. At $\rho = 1.30$ the measured delay falls well *below* the extrapolation of the $[0.85, 1.15]$ fit (at

$p = 41$, 3.9×10^4 steps versus a predicted 1.0×10^5 , a ratio of 0.39; at $p = 59$, ratio 0.71), so the growth is sub-exponential beyond $\rho \approx 1.15$ — the exponential law has an upper edge, and we report it as holding over $\rho \in [0.85, 1.15]$ rather than indefinitely. These high- ρ cells are run at a 160,000-step budget and are *not* censored: all 12/12 seeds grok, with the slowest seed ($\leq 4.7 \times 10^4$ steps) finishing more than threefold below the budget, so the measured delays are full distributions rather than budget-limited lower bounds. At the opposite end, the smallest task ($p = 31$) clamped *below* its critical norm ($\rho = 0.85$, held $\|W\| = 34$) does not grok within budget — it memorizes but its test accuracy creeps up only slowly — suggesting a minimum functional norm below which the solution cannot form efficiently at small p ; at larger p the same $\rho = 0.85$ lies above this floor and accelerates grokking as expected. Both edges sharpen rather than weaken the central finding: across the regime where a single weight norm cleanly sets the function scale, the grokking delay obeys one exponential scaling law in the relative norm.

Relation to delay theory. First-passage / norm-separation accounts of grokking [28] model free training as a crossing problem: weight decay contracts the parameter norm $V_t = \|\theta_t\|^2$ exponentially until it reaches an architecture-dependent threshold V_* , at which the validation transition occurs, giving a delay that is *logarithmic* in the norm ratio, $T_{\text{grok}} - T_{\text{mem}} \approx (2\kappa\eta\lambda)^{-1} \log(V_{\text{mem}}/V_*)$. Our clamp does not measure that logarithmic contraction law — it is a different experiment. By pinning the norm we *block* the contraction route and ask how long grokking takes when the norm cannot reach V_* from above; this is a continuous-dose version of the norm-freeze causal intervention those accounts use to establish necessity (freezing the norm prevents grokking). Our dose-response confirms the central claim of that picture — that the norm is the operative crossing variable — and adds a quantitative law for the pinned regime: the delay grows *exponentially* in the pinned relative norm, steeply enough that a fully frozen high norm reproduces the “never groks within budget” of the freeze intervention. The exponential pinned-norm law is thus complementary to, not the same as, the logarithmic free-contraction law: it characterises the regime where the norm-contraction route is unavailable and generalization must instead be reached at fixed norm. We do not claim a quantitative match between the measured exponent $\alpha \approx 7.5$ and a closed-form theoretical prediction; deriving the pinned-norm exponent from the joint norm-angle dynamics is left to future work.

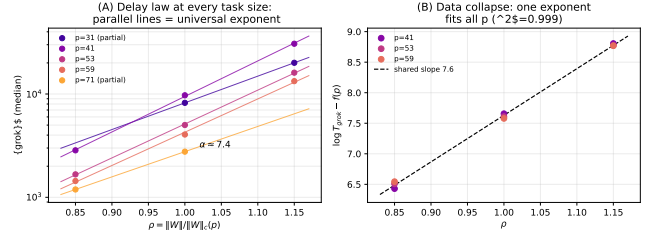


Figure 5: **A scaling law for the grokking delay.** (A) Across task sizes p , holding the norm at $\rho \|W\|_{c(p)}$ delays grokking exponentially over $\rho \in [0.85, 1.15]$ (lines; near-parallel $\Rightarrow$ shared exponent $\alpha \approx 7.5$). The $\times$ markers at $\rho = 1.30$ fall below the extrapolation: the law saturates beyond $\rho \approx 1.15$. (B) Data collapse: with one shared exponent and a per- p offset $f(p)$, all task sizes fall on a single line ($\alpha = 7.49$, $R^2 = 0.996$).

7 Architecture Dependence and the Role of Normalization

Does the critical-norm phenomenon recur beyond the MLP? We test it on a second architecture — a one-layer attention model — in two variants, and find that the answer turns on a single architectural choice, normalization. We use this to map where the norm controls the timescale rather than to claim broad universality.

The functionally relevant norm replicates the threshold. On a one-layer attention model *with* LayerNorm, the *total* weight norm at grokking is not concentrated: it drifts with the training fraction and learning rate (CV 0.15–0.17 across α). This is an artifact of the wrong observable. LayerNorm renormalizes the activations, so the scale of the embedding, attention, and MLP weights (which sit before a normalization) is functionally inert and free to drift. The one norm that sets the function is the *unembedding* U (after the final LayerNorm, it sets the logit scale). Measured on $\|U\|$, the threshold returns: $\|U\|$ at grokking is concentrated (CV 0.03–0.05 across α , 0.04 across a tenfold r range at fixed λ), is set by weight decay, and scales with system size as $\|U\|_c \propto p^{0.77}$ (Fig. 7). The critical-norm threshold thus replicates on the Transformer, but on the functionally relevant norm.

The norm’s control of the timescale is broken by LayerNorm and restored without it. We then clamp by group. Clamping any single group on the LayerNorm model — including $\|U\|$ — moves grokking only weakly and non-specifically (holding $\|U\|$ above its grokking value delays grokking up to $\sim 2\times$, and clamping an upstream group has a comparable effect), because

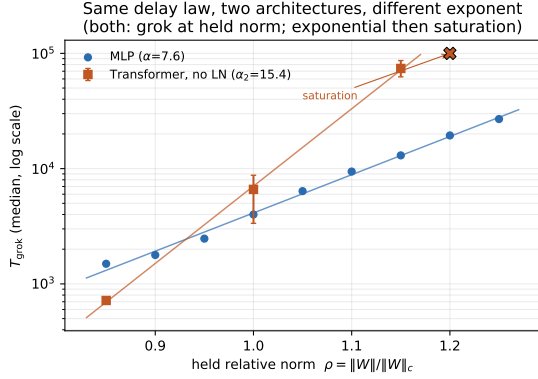


Figure 6: **The delay law transfers to a second architecture with its own exponent.** On the un-normalized one-layer attention model (orange, 16 seeds, bootstrap 95% CIs), the grokking delay is exponential in the held relative norm, $T_{\text{grok}} \propto e^{\alpha_2 \rho}$ with $\alpha_2 = 15.45$ ($R^2 = 0.999$ over $[0.85, 1.15]$), saturating at $\rho = 1.20$ — the same shape as the MLP (blue, $\alpha \approx 7.6$) but $\approx 2\times$ steeper. Both architectures grok at the held norm.

the residual stream mixes contributions and LayerNorm rescales them, so no single weight norm cleanly sets the function. Removing LayerNorm changes this entirely. On the *un-normalized* one-layer attention model, the total weight norm again sets the function scale, and its control of the grokking timescale reappears as a full delay law (Fig. 6). With the critical norm measured at $\|W\|_c \approx 37$ from a free control, an extended-budget clamp sweep groks at *exactly* the held norm in every cell (norm at grokking = $\rho \|W\|_c$: 31.4, 36.9, 42.4, 44.3 for $\rho = 0.85, 1.00, 1.15, 1.20$), all 16/16 seeds and uncensored, with median $T_{\text{grok}} = 718, 6608, 73966, 99920$. Crucially, the $\rho = 1.15$ case — which under a short 30,000-step budget plateaued below 0.9 and looked like *prevention* — groks at $\approx 7.4 \times 10^4$ steps once the budget is extended, confirming a delay rather than a hard block. The delay is exponential in the held norm, $T_{\text{grok}} \propto e^{\alpha_2 \rho}$ with $\alpha_2 = 15.45$ (95% CI $[14.84, 15.98]$, $R^2 = 0.999$ over $\rho \in [0.85, 1.15]$), and saturates beyond: at $\rho = 1.20$ the measured delay is 0.64 $\times$ the exponential extrapolation, the same exponential-then-saturation shape seen on the MLP (§5–6). The exponent is about twice the MLP’s ($\alpha \approx 7.5$): the *form* of the delay law — grok at the held norm, exponential rise, high-norm saturation — transfers across architectures, while its *rate* is architecture-specific. The dose-response is sharp and bidirectional, mirroring the MLP.

Interpretation. The weight norm controls the grokking timescale whenever it sets the scale of the function — on the MLP and on the un-normalized atten-

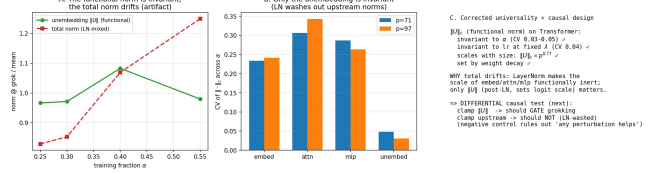


Figure 7: **On a LayerNorm Transformer, the threshold replicates on the functional norm.** (A) The unembedding norm $\|U\|$ at grokking is invariant to the training fraction while the total norm drifts. (B) Across- α coefficient of variation by parameter group: only the unembedding is concentrated (LayerNorm washes out the upstream norms).

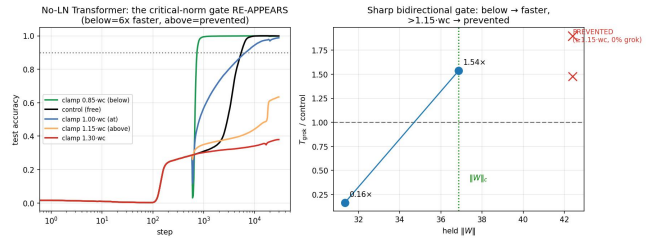


Figure 8: **Without LayerNorm, the norm’s control of the timescale reappears.** (A) Test-accuracy trajectories on the un-normalized attention model: held below $\|W\|_c$ groks 6 $\times$ faster, control intermediate, held above $\|W\|_c$ delayed (trajectories shown at the original 30,000-step budget). (B) Sharp bidirectional dose-response. At extended budget every held-norm case groks at the held norm and follows the exponential delay law of Fig. 6; the above-norm plateaus here are its finite-budget tail, not prevention.

tion model — and LayerNorm is precisely the operation that breaks this link by renormalizing activations and decoupling the weight-norm scale from the function. This shows the causal mechanism *recurs* on a second, un-normalized architecture, and delimits where it does not apply (normalized networks), where the concentrated norm survives only on the post-normalization readout. We therefore speak of structural recurrence under an explicit condition — that a single weight norm sets the function scale — rather than of architecture-independent universality. A methodological corollary: in a normalized network, “the weight norm” must mean the functionally relevant post-normalization norm, not the raw total.

8 Generality: a Non-Fourier Task

Modular arithmetic has cyclic, Fourier structure, which raises the question of whether the critical norm is an artifact of that structure. We test on k -sparse parity — the canonical non-Fourier grokking task [4], where the label is the product of k relevant bits among n and the solution is combinatorial feature selection rather than a Fourier circuit — using an MLP (no normalization, no embedding), so the total weight norm is again the functionally relevant norm.

Two parts of the phenomenon transfer. First, grokking occurs. Second, the grokking norm is again a concentrated, weight-decay-set, learning-rate-invariant target: at a fixed task and weight decay, varying the learning rate over a fourfold range changes the grokking norm by only $\approx 3\%$ while the grokking time varies $\sim 4\times$, and weight decay sets the grokking norm along a clean monotone frontier (as on modular arithmetic). What does *not* transfer is the regularization-unreachable above-norm delay. The grokking norm here drifts with the amount of training data (it is not data-invariant), and the norm clamp lies *on* the weight-decay frontier in both directions: unlike the modular case, the high-norm regime is reachable by reducing weight decay, so clamping the norm high reproduces what weak regularization already does, rather than holding the network in a slow, high-norm state that no weight decay can reach. A plausible reason is structural: sparse parity admits many generalizing solutions spread over a range of norms, whereas modular addition is solved by a near-unique Fourier construction pinned to a particular scale, so only the latter has a high-norm regime that regularization cannot otherwise enter (we offer this as a hypothesis, not a tested claim).

We therefore distinguish two claims by their scope. The *concentrated, regularization-set grokking norm* is task-general (modular arithmetic and sparse parity; MLP and, on the functional norm, the Transformer). The *regularization-unreachable above-norm delay* — the slow, high-norm states the clamp holds that no weight decay reaches, together with the initialization washout — is specific to settings where a single weight norm sets the function scale (modular arithmetic and the unnormalized attention model). Table 2 summarizes.

9 Discussion

Throughout, we keep claims separated by evidence strength — what is observationally corroborated, what is established causally, and what holds only over the tested finite range — as summarized in Table 3.

Reconciling the threshold and “norm-is-not-causal” accounts. Our central result dissolves a stand-

Table 2: Which properties transfer across task and architecture.

property	mod. add.	parity
grokking occurs	yes	yes
norm at grok set by weight decay	yes	yes
norm at grok invariant to learning rate	yes	yes
norm at grok invariant to data	yes	no
reg.-unreachable above-norm delay	yes	no

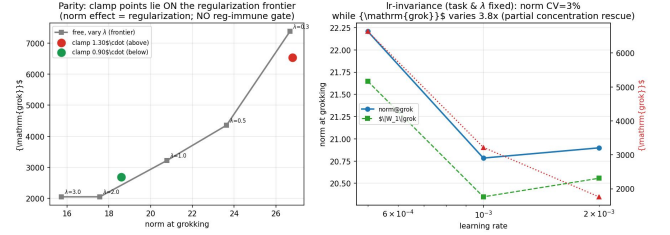


Figure 9: **Sparse parity (non-Fourier).** (Left) The norm clamps lie on the weight-decay frontier in both directions (no regularization-unreachable high-norm state, unlike modular arithmetic). (Right) At fixed task and weight decay, the grokking norm is nearly invariant to learning rate ($\approx 3\%$) while the grokking time varies $\sim 4\times$ — a concentrated, regularization-set target.

ing disagreement. One line reports a concentrated weight-norm threshold for grokking [6, 18, 19]; another argues the norm is correlative or not the operative variable [16, 17, 20, 21]. The clamp matrix shows both are seeing facets of a *rate* law: the network groks at whatever norm it is held at, and the held norm sets the grokking time exponentially (2). The concentrated $\|W\|_c$ seen under free training is therefore the norm the weight-decay relaxation *reaches* on its natural timescale — which is why it looks like a threshold — and not a value grokking requires, which is why grokking is observed above and below it. This complements norm-separation delay theory [28], which predicts the *free-training* delay as logarithmic in the norm ratio from an exponential norm-contraction argument: that theory explains why free training groks on a finite timescale (the norm contracts to threshold), while our pinned-norm intervention — a continuous-dose form of its norm-freeze causal test — isolates the norm as the operative variable and measures the (exponential) cost of being held away from threshold.

A bridge between two accounts. The weight-norm and circuit accounts appear as levels of one transition: weight decay drives the norm down, and fea-

Table 3: Claims and evidence strength. We separate what is corroborated, what is causal, and what is scoped, to make the reach of each claim explicit.

claim	evidence	status
critical norm under free dynamics	observational sweeps (§4)	corroborated
norm controls grokking time	matched clamp matrix (§5)	causal
exponential delay over $\rho \in [0.85, 1.25]$	dense 10-level ρ sweep, $R^2=0.994$	established (MLP)
shared exponent across p	multi- p collapse, $R^2=0.996$	finite-range scaling
saturation beyond $\rho \approx 1.25$	$\rho=1.30$ cells, uncensored	observed
norm vs. projection	$\rho=1.00$ control	causal
second-architecture delay law	un-normalized attention, $\alpha_2 \approx 15$, $R^2=0.999$	causal (confirmed)
sparse-parity transfer	non-Fourier MLP	partial
relation to delay theory	complements log contraction law	qualitative

ture formation — and generalization — follow on the timescale the norm sets. The intervention shows the norm causally sets the timescale; the temporal alignment suggests, but does not prove, that it organizes the features.

Normalization decouples the norm from the function. The architecture comparison (§7) gives a mechanistic account of *when* the picture applies. In an un-normalized network the weight norm sets the function scale, and its control of the timescale is sharp; LayerNorm renormalizes the activations, so the scale of upstream weights no longer affects the function and no single weight norm cleanly controls the transition. The concentrated norm survives on a LayerNorm model only on the post-normalization readout (the unembedding). We thus read the weight-norm account as conditional on the norm being functionally meaningful — a condition normalization layers are designed to relax.

Contrast with double descent. Epoch-wise double descent is a related but mechanistically distinct transition: in a multi-task linear model it is a saddle-node bifurcation with transition-time exponent $\delta = 1/2$, whereas the grokking time here grows much more steeply with the control parameter. We draw the contrast at the level of mechanism (a saddle-node escape versus a relaxation-set timescale) rather than asserting distinct universality classes from exponents, given the limited size range.

Generality and its limits. Across a non-Fourier task (sparse parity) and a second architecture (attention), the concentrated, weight-decay-set, learning-rate-invariant grokking norm recurs, while the regularization-unreachable above-norm delay (slow high-norm states no weight decay reaches; initialization washout) is confined to settings where a single weight norm sets the function scale. We therefore make the broad claim only for the concentrated grokking norm and the delay law, and scope the regularization-unreachable part explicitly.

Is there an upper norm above which grokking never occurs? Within the surveyed range $\rho \in [0.85, 1.30]$ grokking always occurs — it is only delayed — and the delay grows *sub*-exponentially at the top of this range rather than diverging, so we see no evidence of a finite norm at which the delay becomes infinite. We cannot, however, rule out an upper threshold ρ^* at norms well above those tested: our intervention establishes a delay law and its saturation, not the absence of a hard ceiling at extreme norm. We leave the existence of such a ρ^* open.

Is it the norm, or the projection? Because the clamp re-projects the weights every step, one might worry the repeated projection itself — rather than the norm value — causes the delay. The $\rho = 1.00$ arm is the control for this: it applies the identical per-step projection but at the network’s natural norm, and it groks on essentially the free-training timescale (no delay). Delay appears only as the projected norm departs from $\|W\|_c$, and scales with that departure, so the operative variable is the held norm value, not the act of projecting.

Limitations. The critical norm depends on weight decay and system size, so its concentration is one of structure within a fixed family, not of an absolute value. The clamp is a strong, sustained constraint rather than a subtle perturbation (it changes magnitude while preserving direction). The exponential delay law is established on the MLP over $\rho \in [0.85, 1.25]$ and, at an extended budget, on the un-normalized attention model over $\rho \in [0.85, 1.15]$ ($\alpha_2 = 15.45$, $R^2 = 0.999$), each saturating beyond; we did not extend the sweep to a third architecture. The architecture study uses one-layer attention models. On the LayerNorm model we establish that no single weight norm controls the timescale but do not exhaustively rule out a multi-norm account. The scaling exponents are measured over a modest size range and reported as suggestive.

10 Conclusion

The weight norm causally sets the timescale of grokking. Under free training the norm at grokking is sharply concentrated at $\|W\|_c$ (in agreement with recent threshold reports), set by weight decay and scaling with system size. But a matched-counterfactual clamp shows this is not a hard threshold: the network groks at whatever norm it is held at, and the time to grok grows as a clean exponential in the held norm ($T_{\text{grok}} \propto e^{\alpha\rho}$, $R^2 = 0.99$). Holding the norm above $\|W\|_c$ therefore delays grokking rather than preventing it — an apparent “prevention” under a fixed budget is the finite-budget tail of this law — and the norm dominates the grokking time ($\approx 19\times$) over the learning rate ($\approx 2\times$), with the two dissociable. This reconciles the threshold and “norm-is-not-causal” accounts: the concentrated $\|W\|_c$ is the norm the relaxation reaches on its natural timescale, not a value grokking requires, so grokking is seen above and below it. The picture recurs on an un-normalized attention model and is altered by LayerNorm, which decouples the weight-norm scale from the function; on sparse parity the concentrated, weight-decay-set, learning-rate-invariant grokking norm recurs. The result is a causal complement to norm-separation delay theory — whose closed-form delay is logarithmic in the norm ratio under free contraction — and recasts grokking as a transition whose timescale is set by the relaxation of a functionally meaningful weight norm. Code and data: <https://github.com/ClevixLab/critical-norm-grokking>. We release the runners and analyzers, and the per-seed metrics for the full $\rho \times \eta$ clamp matrix, the multi- p scan, and the dense ρ sweep, so that every fit, table, and figure here can be reproduced end to end.

Data availability

The code and the per-seed metrics required to reproduce every number, table, and figure in this paper are openly available at <https://github.com/ClevixLab/critical-norm-grokking>. The complete raw dataset, including the full per-step structural snapshots, is archived on Zenodo (DOI to be inserted at proof stage).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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A Reproducibility Checklist

All results are produced by the released code (<https://github.com/ClevixLab/critical-norm-grokking>). Each runner writes a timestamped output directory containing its own script, per-seed metrics (`T_grok` for every seed, per-layer norms, norm-at-grokking), and a checkpoint enabling resume-after-interruption; analyzers read these directories and emit the numbers quoted here. The mapping from artifact to result is:

- **Observational critical norm** (Fig. 1, §4): free-training sweeps over training fraction and learning rate; the per- p controls also supply $\|W\|_c(p)$.
- **Causal clamp matrix** (Fig. 2, Table 1, §5): `run_efflr_dissoc` — the $\rho \times \eta$ matrix at $p = 59$, 16 seeds, with the matched control.
- **Dense ρ sweep** (Fig. 3, §5): `run_denserho / analyze_denserho` — ten ρ levels at $p = 59$, giving $\alpha = 7.64$, $R^2 = 0.994$.
- **Scaling collapse** (Fig. 5, §6): `run_delaylaw_pscan / analyze_delaylaw_pscan` — the multi- p scan. The single-exponent verdict (Table 3) is reproduced by `analyze_scaling_variable` (per- p α bootstrap CIs, shared-vs-per- p model comparison, normalisation-variable test).
- **Second architecture** (Fig. 6, §7): `run_tfnoln_long / analyze_tfnoln_long` — the extended-budget un-normalized transformer, giving $\alpha_2 = 15.45$, $R^2 = 0.999$.
- **LayerNorm functional norm and sparse parity** (Figs. 7, 8, 9; Table 2): the LayerNorm and parity runners in the same repository.

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